

UNIVERSITY OF THE ARMED FORCES - ESPE

Computer Science Department

Object Oriented Programming

36863

Homework08 - MethodDefinitionOnChickenFarmApp

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```java
public void add(Chicken chicken) {
      if (chicken == null) {
    throw new IllegalArgumentException("Chicken cannot be null.");
}

for (Chicken currentChicken : chickens) {
    if (currentChicken != null
            && currentChicken.getId() == chicken.getId()) {
        throw new IllegalArgumentException(
            "A chicken with that ID already exists."
        );
    }
}

for (int i = 0; i < chickens.length; i++) {
    if (chickens[i] == null) {
        chickens[i] = chicken;
        return;
    }
}

throw new IllegalStateException("The coop is full.");
}

public void remove(int chickenId) {
    if (chickenId <= 0) {
    throw new IllegalArgumentException(
        "Chicken ID must be greater than zero."
    );
}

for (int i = 0; i < chickens.length; i++) {
    if (chickens[i] != null
            && chickens[i].getId() == chickenId) {
        for (int j = i; j < chickens.length - 1; j++) {
            chickens[j] = chickens[j + 1];
        }

        chickens[chickens.length - 1] = null;
        resetIteration();
        return;
    }
}
```

```
for (int i = 0; i < chickens.length; i++) {
    if (chickens[i] != null
            && chickens[i].getId() == chickenId) {
        for (int j = i; j < chickens.length - 1; j++) {
            chickens[j] = chickens[j + 1];
        }

        chickens[chickens.length - 1] = null;
        resetIteration();
        return;
    }
}
throw new IllegalArgumentException(
    "No chicken found with that ID."
);
}

public void resetIteration() {
     currentIndex = 0;

}

public void next() {
while (currentIndex < chickens.length
        && chickens[currentIndex] == null) {
    currentIndex++;
}

if (currentIndex >= chickens.length) {
    System.out.println("No more chickens.");
    return;
}

System.out.println(chickens[currentIndex]);
currentIndex++;
}
```

```
--- Anthony Arevalo's Chicken Farm App ---
1. Add chicken
2. Remove chicken
3. Reset chicken iteration
4. Next chicken
5. Show chicken count and list
6. Exit
Choose an option: 1
Chicken ID: 02
Name: Anthony
Color: Blanck
Date of birth (YYYY-MM-DD): 2024-02-23
Is the chicken molting? (y/n): y
Chicken added.

--- Anthony Arevalo's Chicken Farm App ---
1. Add chicken
2. Remove chicken
3. Reset chicken iteration
4. Next chicken
5. Show chicken count and list
6. Exit
Choose an option: 1
Chicken ID: 03
Name: Alex
Color: red
Date of birth (YYYY-MM-DD): 2023-05-21
Is the chicken molting? (y/n): y
Chicken added.

--- Anthony Arevalo's Chicken Farm App ---
1. Add chicken
2. Remove chicken
3. Reset chicken iteration
4. Next chicken
5. Show chicken count and list
6. Exit
Choose an option: 5
Number of chickens in the coop: 3
Chicken{id=1, name=Demain, color=pink, bornOnDate=Sat Jun 20 00:00:00 GMT-05:00 2020, age=6, molting=true}
Chicken{id=2, name=Anthony, color=Blanck, bornOnDate=Fri Feb 23 00:00:00 GMT-05:00 2024, age=2, molting=true}
Chicken{id=3, name=Alex, color=red, bornOnDate=Sun May 21 00:00:00 GMT-05:00 2023, age=3, molting=true}
```